

Automated Annotation of Complex Natural Products Using A Modular Fragmentation-based Structure Assembly (MFSA) Strategy

User Manual of CNPs-MFSA

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Introduction

Complex natural products (CNPs) Structure Annotation Tool is a user-friendly application for processing and analyzing LC-MS spectra. It provides functionalities to recognize target natural products, calculate molecular compositions, determine possible adduct ions, identify characteristic neutral losses, extract feature ions, calculate cosine similarity scores between standard and query spectra, and provide possible structure annotation. The tool is designed for both beginners and experts in natural product chemistry, offering an intuitive interface to streamline natural product analysis tasks while maintaining the flexibility for customization.

Installation and Setup (On PC)

1. Application Requirements:

- Python 3.8 or higher. (Recommend Python 3.11)
- Required Python libraries:

a. Data processing and file handling:

`pandas, numpy`

b. Chemical analysis and mass spectrometry data handling:

`sqlite3, pyteomics, matchms, rdkit`

c. Building and managing a web-based dashboard using Dash:

`dash, dash_bootstrap_components, plotly, flask, chardet`

2. Installation Steps:

- Download and install Pycharm with Python (Recommend) or other IDE like Anaconda
- Clone the repository:

```
git clone <https://github.com/MiZhang47/CNPs-MFSA.git >
```

- Navigate to the directory: `CNPs-MFSA`

3. Running the Application:

- Start the application by executing: `CNPs-MFSA_GUI.py`
- Open a web browser and navigate to: <http://127.0.0.1:8050>

Using the Tool

1. Tabs Overview

1.1. Target Recognition: Extract specific target precursor ion from uploaded data.

**CNPs-MFSA**

Welcome! Complex natural products (CNPs) Structure Annotation Tool was constructed based on a modular fragmentation-based structural assembly (MFSA) strategy. This tool provides functionalities to recognize target natural products, calculate molecular compositions, determine possible adduct ions, identify characteristic neutral losses, extract feature ions, calculate cosine similarity scores between standard and query spectra, and provide possible structure annotation. Please select the relevant tab to proceed.

1. Target Recognize	2. Composition Calculate	3. Adduct Ion Calculate	4. Neutral Loss Extract	5. Feature Ion Extract	6. Cosine Score Calculate	7. Annotation	8. Visualization
---------------------	--------------------------	-------------------------	-------------------------	------------------------	---------------------------	---------------	------------------

Import MGF file (multiple files):

Drop or Select MGF File

No MGF file uploaded

Type Name 1:

Type 1 Feature Formulas:

Input Type feature ion formulas

Mass Range of Product Ions (m/z): to

Charge: Mass Tolerance (ppm): Hit Score:

No recognition performed. Please upload both CSV and MGF files.

Import CSV file (multiple files):

Drop or Select CSV File

No CSV file uploaded

1.2. Composition Calculation: Analyze molecular composition from spectra.

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Import query MS Spectra (multiple files):

Drop or Select CSV Files

No CSV file uploaded

Charge: Mass Tolerance (ppm):

Select file to view results:

No results available

Import formula database (POS/NEG):

Drop or Select CSV Files

No DB file uploaded

1.3. Adduct Ion Calculation: Identify possible adduct ions in mass spectra.

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1. Target Recognize

2. Composition Calculate

3. Adduct Ion Calculate

4. Neutral Loss Extract

5. Feature Ion Extract

6. Cosine Score Calculate

7. Annotation

8. Visualization

Import query MS Spectra (multiple files):

Drop or Select CSV Files

No CSV file uploaded

Charge: Mass Tolerance (ppm):

Calculate

Download All Results

Select file to view results:

Select a file

Alternative: Import known compound MS database (POS/NEG):

Drop or Select DB Files (.csv)

No database file uploaded.

1.4. Neutral Loss Labeling: Identify neutral loss patterns in spectra.

**CNPs-MFSA**

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1. Target Recognize

2. Composition Calculate

3. Adduct Ion Calculate

4. Neutral Loss Extract

5. Feature Ion Extract

6. Cosine Score Calculate

7. Annotation

8. Visualization

Import MGF file (multiple files):

Drop or Select MGF File

No MGF file uploaded

Charge: Mass Tolerance (ppm):

Loss Name 1:

Target Loss 1:

Input target loss formulas

Carbon Range 1 From : to

Carbon Range 2 From : to

✖

+ Add New Target Loss

Run Neutral Loss Extraction

Download All Results

Click the button to start the extraction.

Import CSV file (multiple files):

Drop or Select CSV File

No CSV file uploaded

Import formula database (POS/NEG):

Drop or Select CSV Files

No DB file uploaded

<>

1.5. Feature Ion Labeling: Extract and analyze feature ions.

**CNPs-MFSA**

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1. Target Recognize	2. Composition Calculate	3. Adduct Ion Calculate	4. Neutral Loss Extract	5. Feature Ion Extract	6. Cosine Score Calculate	7. Annotation	8. Visualization
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Import MGF file (multiple files):

Drop or Select MGF File

No MGF file uploaded

Import CSV file (multiple files):

Drop or Select CSV File

No CSV file uploaded

Ion Name 1:

Specific Ion Set 1 :

Input feature ion formulas

Mass Range of Product Ions (m/z): to

Charge: Mass Tolerance (ppm): Intensity Normalization: Top Number:

[Download All Results](#)

Select file to view results:

1.6. Cosine Score Calculation: Calculate cosine similarity scores for spectral matching.

**CNPs-MFSA**

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1. Target Recognize	2. Composition Calculate	3. Adduct Ion Calculate	4. Neutral Loss Extract	5. Feature Ion Extract	6. Cosine Score Calculate	7. Annotation	8. Visualization
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Import Query MGF file (multiple files):

Drop or Select MGF File

No MGF file uploaded

Import Query CSV file (multiple files):

Drop or Select CSV File

No CSV file uploaded

Import Standard MGF file (multiple files):

Drop or Select MGF File

No MGF file uploaded

Import Standard CSV file (multiple files):

Drop or Select CSV File

No CSV file uploaded

Mass Range of Product Ions (m/z): to

Intensity Normalization: Cosine:

[Download All Results](#)

Select file to view results:

Please select a file to view results.

1.7. Annotation for daphnane (Alternative)

1. Target Recognize	2. Composition Calculate	3. Adduct Ion Calculate	4. Neutral Loss Extract	5. Feature Ion Extract	6. Cosine Score Calculate	7. Annotation	8. Visualization
Import Query CSV file (multiple files): Drop or Select CSV File Uploaded CSV files: Thyme56-centroid-MD_quant-re-mf-ad-nl-fi.csv Run Annotation Download All Results							
Import pseudo-DB file: Drop or Select DB File Uploaded pseudo-DB file: pseudo-MD.db							
Import Substituent CSV file: Drop or Select CSV File Uploaded CSV files: Acyl.csv							
Click Run Annotation to start processing.							

1.8. Visualization

**CNPs-MFSA**

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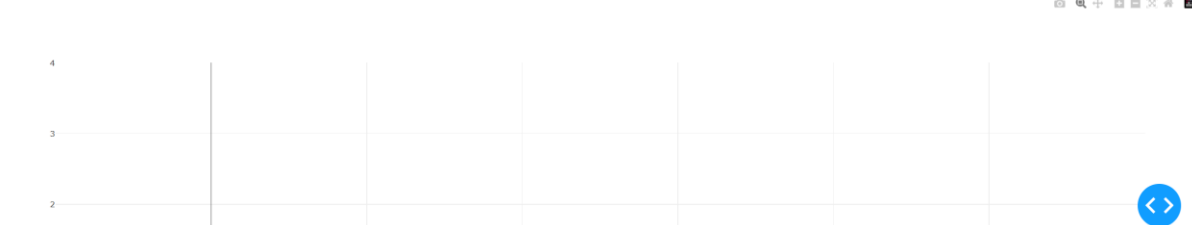
1. Target Recognize	2. Composition Calculate	3. Adduct Ion Calculate	4. Neutral Loss Extract	5. Feature Ion Extract	6. Cosine Score Calculate	7. Annotation	8. Visualization
---------------------	--------------------------	-------------------------	-------------------------	------------------------	---------------------------	---------------	------------------

Import Query CSV file (multiple files):

Drop or Select CSV File

No CSV file uploaded

[Run Visualization](#)



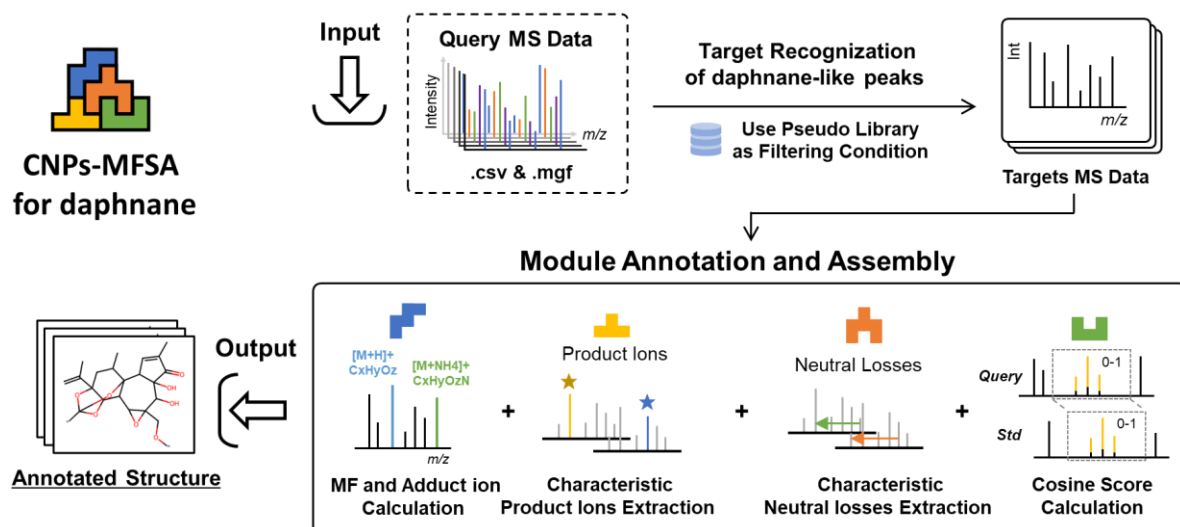


Figure 1. Automatic Annotation Workflow of Thymelaeaceae Daphnanes Using Modular Fragmentation-based Structure Assembly (MFSA) Strategy.

2. Example Workflow for Daphnane-type Diterpenoids

Step 1: Target Recognize

(1) Upload Files: Select MGF and CSV files processed by MZmine

***Note:** In order to speed up the process, we recommended to process raw files as **centroid** data. Before uploading, all **.mgf** files generated by MZmine need to be opened with Notepad, and all **FEATURE_ID** need to be changed to **TITLE** using the search and replace mode.

<pre>BEGIN IONS FEATURE_ID=1 RTINSECONDS=293.536 CHARGE=1+ MSLEVEL=2 MERGED_STATS=1 / 20</pre>	→	<pre>BEGIN IONS TITLE=1 RTINSECONDS=293.536 CHARGE=1+ MSLEVEL=2 MERGED_STATS=1 / 20</pre>
--	---	---

(2) Define Parameters: Set up type number, type name, mass range, charge and feature formulas (characteristic product ions).

```
1486 # Detect and extract spectra from an MGF file based on provided criteria
1487 def detect_recognize_spectrum(filename: str, feature_formulas: list, min_mass: float, charge: int, tolerance_ppm: float,
1488 > hit_score: int, max_mass_option: str, max_mass_value: float = None):...
```

```
1531 if max_mass < 300 or max_mass > 1500:
1532     continue
1533
1534 # Filter product ions based on mass range
1535 product_ions = spectrum['m/z array']
1536 intensity = spectrum['intensity array']
1537 ind = np.nonzero((product_ions >= min_mass) & (product_ions <= max_mass))[0]
1538 print(f"Number of product ions within mass range: {len(ind)}")
1539
1540 if len(ind) == 0:
1541     print("No product ions found within specified mass range. Skipping spectrum.")
1542     continue
1543
1544 # Normalize intensities for scoring
1545 intensity_max = np.max(intensity[ind])
1546 intensity_min = np.min(intensity[ind])
1547 normalized_intensity = (intensity[ind] - intensity_min) / (intensity_max - intensity_min)
1548 nor_intensity = np.zeros_like(intensity)
1549 nor_intensity[ind] = normalized_intensity
1550 print(f"Normalized intensities for spectrum.")
1551
1552 # Count hits based on the provided formulas and tolerance
1553 hits = 0
1554 for intensity_value, product_ion in zip(nor_intensity, product_ions):
1555     if intensity_value > 0.05:
1556         for formula in feature_formulas:
1557             mz_min, mz_max = recognize_mz_range(formula, charge, tolerance_ppm)
1558             if mz_min <= product_ion <= mz_max:
1559                 hits += 1
1560                 break # Stop checking further formulas once a hit is found for this product ion
1561 print(f"Hits found for spectrum: {hits}")
```

Figure 2. Parameters of Tab 1 for daphnane annotation.

As for daphnane-type diterpenoids from Thymelaeaceae, the parameters were set as following:

- Type Name 1: **D** [Abbreviation for normal daphnane-type diterpenoids]
- Type 1 Feature Formulas: (The following ion formula were composed of the measured product ion formula of standard normal daphnanes within the mass range and the predicted product ion formula in the pseudo-daphnane library without duplication.)

$C_{20}H_{28}O_4$, $C_{20}H_{28}O_5$, ... $C_{18}H_{16}O_2$, $C_{18}H_{16}O_1$, $C_{18}H_{16}$, $C_{17}H_{18}O_1$, $C_{17}H_{16}O_1$, $C_{15}H_{18}O_2$

***Note:** For detailed ion information, see the [feature ion and neutral loss.txt](#) file, which can be find in GitHub repository [CNPs-MFSA](#).

- Mass Range of Product Ions (m/z): **230** to **400** (Considering that the product ions derived from certain aglycones may be affected by the intensity of the substituent ions, resulting in a decrease in relative intensity, it is possible to normalize the product ions within a specific range to search for specific product ions.)
- Charge: **+1** (positive ion mode)
- Mass Tolerance (ppm): **5** (Q-Exactive hybrid quadrupole Orbitrap mass spectrometer)
- Hit Score: **6** (The minimum score threshold for normal daphnane-type diterpenoids)
- Type Name 2: **MD** [Abbreviation for macrocyclic daphnane orthoesters]
- Type 2 Feature Formulas: (The following ion formula were composed of the measured product ion formula of standard macrocyclic daphnane orthoesters within the mass range and the predicted product ion formula in the pseudo-daphnane library without duplication.)

$C_{30}H_{44}O_8$, $C_{30}H_{44}O_9$, ... $C_{30}H_{46}O_8$, $C_{28}H_{34}O_9$, $C_{32}H_{38}O_5$, $C_{27}H_{30}O_2$, $C_{32}H_{38}O_2$, $C_{27}H_{32}O_6$

***Note:** For detailed ion information, see the [feature ion and neutral loss.txt](#) file, which can be find in GitHub repository [CNPs-MFSA](#).

- Mass Range of Product Ions (m/z): **350** to **Use Precursor Ion** (same as above)
- Charge: **+1** (positive ion mode)
- Mass Tolerance (ppm): **5** (Q-Exactive hybrid quadrupole Orbitrap mass spectrometer)
- Hit Score: **6** (The minimum score threshold for normal daphnane-type diterpenoids)

***Note:** On a personal computer, the processing speed of **centroid data** is about **120 precursor ions/minute**, so it takes 100 minutes to process a test data containing about 12,000 precursor ions; while the processing speed of **profile data** is about **60 precursor ions/minute**, it takes 200 minutes to complete the processing of the same test data.

(4) Download Results: download the generated .zip file and the results were shown below with new rows, namely **type score** and **type**. The **xxx-re.csv** was used as the **input file** and switched to the next tab "**Composition Calculate**". The **xxx-re.mgf** was used as the **input file** of Step 4, 5, and 6.

Thyme56-centroid.mgf
Thyme56-centroid-re.mgf
Thyme56-centroid_quant.csv
Thyme56-centroid_quant-re.csv

	A	B	C	U	V	W	X	Y	Z	AA	AB	AC	AD	AE	AF	AG	AH	BH	BI	BJ
1	row ID	row m/z	row retent	Dod-fr Pei	Dod-b Pei	Dod-l Pea	Dod-r Pea	Dod-s Pei	Dje-fr Pea	Dje-ft Pea	Dge-fr Pei	Dps-fr Pei	Dps-ft Pei	Dpo-fr Pei	Dpo-l Pea	Dpo-s Pei	Wdo-w Pei	D Score	MD Score	type
2	1	655.2525	8.734482	0	0	0	0	0	0	8.47E+08	0	0	1.09E+08	0	0	0	0	15	0	D
3	2	591.2585	8.815949	0	0	674910.3	8809730	1687892	0	5.76E+08	0	0	2.44E+09	0	0	0	6439611	20	0	D
4	3	657.2678	8.894676	0	0	0	0	0	0	97400000	0	0	1.18E+08	0	0	0	0	19	0	D
5	5	649.3003	10.66597	0	806474.1	0	1388519	0	0	54400000	7030045	0	3810075	0	0	0	0	16	0	D
6	6	629.2382	7.71742	0	0	0	4278957	5021816	0	58400000	7110380	0	0	0	0	0	0	18	0	D
7	7	647.2847	10.00882	1664660	0	0	0	19400000	0	62600000	0	0	9713001	0	0	0	0	13	0	D
8	8	643.2881	10.84486	0	0	0	0	0	0	1.13E+08	0	0	1.71E+09	0	0	0	0	12	0	D
9	9	603.2223	6.755403	0	0	0	0	4851743	0	44100000	2.54E+08	0	0	0	0	0	0	17	0	D
10	10	637.3366	12.58025	0	8365642	0	1542323	0	0	47400000	0	0	2.65E+08	0	0	0	0	19	0	D

Figure 6. Output results of Step 1: Target Recognize.

Step 2: Composition Calculate

1. Target Recognize	2. Composition Calculate	3. Adduct Ion Calculate	4. Neutral Loss Extract	5. Feature Ion Extract	6. Cosine Score Calculate	7. Annotation	8. Visualization
Import query MS Spectra (multiple files): Drop or Select CSV Files Uploaded 1 CSV file(s): Thyme56-centroid_quant-re.csv		Import formula database (POS/NEG): Drop or Select CSV Files Uploaded DB file: Positive Ion Formula.db					
Charge: 1 Mass Tolerance (ppm): 5							
Calculate Download All Results							

Figure 7. Parameters in Step 2: Adduct Ion Calculate.

(1) Upload CSV Files: Select [xxx-re.csv](#) files processed by **Step 1**.

(2) Upload DB Files: Select [Positive Ion Formula.db](#) files generated by [formula DB generation.py](#), which could be found in Github.

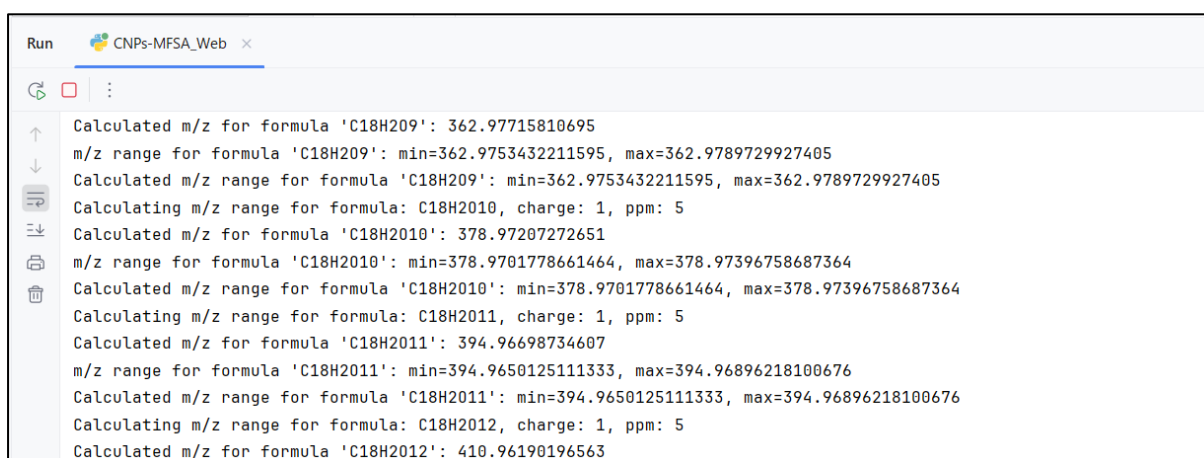
***Note:** [formula DB generation.py](#) generates molecular formulas based on carbon (C), hydrogen (H), and oxygen (O) atoms, as well as Index of Hydrogen Deficiency (IHD) to determine valid formulas, then saves them to an SQLite database. It also simulates the addition of adduct ions, such as $[M+NH_4]^+$ and $[M+HCOO]^-$, to molecules and stores them in separate databases for positive and negative ionization modes. Users could modify [generate_organic_compounds\(\)](#) to include additional elements or change element number, and adjust IHD range for different constraints. Users also could extend [add_additional_groups\(\)](#) to include other adduct ions.

```
14 # Generate a list of organic compounds based on possible combinations of C, H, and O
15 def generate_organic_compounds(): 1 usage
16     compounds = [] # List to store the generated compounds
17     # Iterate through possible numbers of carbon atoms (from 1 to 60)
18     for nC in range(1, 81):
19         # Iterate through possible numbers of hydrogen atoms (from 2 to 80, in steps of 2)
20         for nH in range(2, 91, 2):
21             # Iterate through possible numbers of oxygen atoms (from 1 to 20)
22             for nO in range(1, 26):
23                 IHD = calculate_IHD(nC, nH, nO) # Calculate the IHD for the compound
24                 # Only store compounds with an IHD between 3 and 25 (inclusive)
25                 if 3 <= IHD <= 25:
26                     compound = (nC, nH, nO)
27                     compounds.append(compound)
28     return compounds
29
30 # Simulate the addition of functional groups to a list of compounds
31 def add_additional_groups(compounds): 1 usage
32     # Add an NH3 group to each compound
33     nh3_added = [(nC, nH + 3, nO, 1) for nC, nH, nO in compounds]
34     # Add a CH2O2 group to each compound
35     ch2o2_added = [(nC + 1, nH + 2, nO + 2) for nC, nH, nO in compounds]
36     # Add a CH3NH2 group to each compound
37     ch5n_added = [(nC + 1, nH + 5, nO, 1) for nC, nH, nO in compounds]
38     return nh3_added, ch2o2_added, ch5n_added
```

Figure 8. Parameters of formula DB generation.py

The [Positive Ion Formula.db](#) file provided in the input data was designed for the calculation of daphnane-type diterpenes, with **carbon atom** numbers ranging from **1** to **80**, hydrogen atom numbers ranging from **2** to **90**, oxygen atom numbers ranging from **1** to **26**, and Index of Hydrogen Deficiency (IHD) values ranging from **3** to **25**. Furthermore, in addition to $[M+H]^+$, $[M+NH_4]^+$, and $[M+CH_3NH_2+H]^+$ considered as adduct ion species for positive ion modes.

***Note:** According to the current parameter settings, the element composition can be correctly calculated for peaks with **row m/z** between **400-1300 Da**, but for peaks outside the mass range, about **10%** of the peaks cannot be calculated. Therefore, the author should conduct manual verification and added the molecular formula manually.

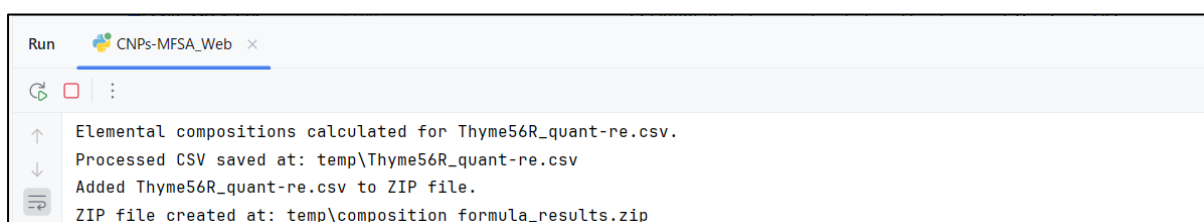


```

Run CNPs-MFSA_Web x
Calculated m/z for formula 'C18H209': 362.97715810695
m/z range for formula 'C18H209': min=362.9753432211595, max=362.9789729927405
Calculated m/z range for formula 'C18H209': min=362.9753432211595, max=362.9789729927405
Calculating m/z range for formula: C18H2010, charge: 1, ppm: 5
Calculated m/z for formula 'C18H2010': 378.97207272651
m/z range for formula 'C18H2010': min=378.9701778661464, max=378.97396758687364
Calculated m/z range for formula 'C18H2010': min=378.9701778661464, max=378.97396758687364
Calculating m/z range for formula: C18H2011, charge: 1, ppm: 5
Calculated m/z for formula 'C18H2011': 394.96698734607
m/z range for formula 'C18H2011': min=394.9650125111333, max=394.96896218100676
Calculated m/z range for formula 'C18H2011': min=394.9650125111333, max=394.96896218100676
Calculating m/z range for formula: C18H2012, charge: 1, ppm: 5
Calculated m/z for formula 'C18H2012': 410.96190196563
  
```

Figure 9. Screenshot of the python console while running Step2: Adduct Ion Calculate.

Calculation was complete with the following information:

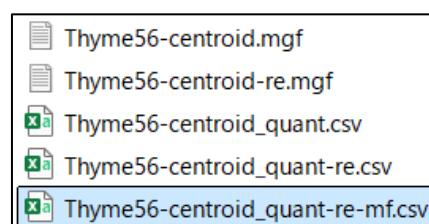


```

Run CNPs-MFSA_Web x
Elemental compositions calculated for Thyme56R_quant-re.csv.
Processed CSV saved at: temp\Thyme56R_quant-re.csv
Added Thyme56R_quant-re.csv to ZIP file.
ZIP file created at: temp\composition formula_results.zip
  
```

Figure 10. Screenshot of the python console while complete Step2: Adduct Ion Calculate.

(5) Download Results: download the generated **xxx.zip** file and the results were shown below with new row, **elemental composition**. The **xxx-mf.csv** was used as the **input file** and switched to the next tab "Adduct Ion Calculate".



	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
1	row ID	row m/z	row retention time	D Score	MD Score	type	elemental composition	20221207, 20221221	20221207	20221105, 20221105	20240421, 20221207, 20221207	20221105, 20240421, 20221105	20221105, 20240421, 20221105					
2	2	543.2222	4.4936056	16	0	D	C29H34O10	4.1E+08	0	0	0	0	0	0	0	0	0	0
3	6	635.321	12.122931	9	0	D	C37H46O9	2.79E+08	22649791	0	2799330	58690113	17138181	0	0	2.5E+08	0	2510044
4	9	605.2379	7.003309033	18	0	D	C34H36O10	84438393	1189480	0	0	0	0	0	12614013	0	0	12450572
5	11	647.285	10.60498556	11	0	D	C37H42O10	1.04E+08	9550283	0	1.8E+08	0	0	0	0	1.24E+09	0	0
6	12	649.3002	10.68500329	12	0	D	C37H44O10	1.04E+08	0	0	0	15626314	5556343	0	12159979	6395130	0	53544545
7	14	603.2223	6.764385931	12	0	D	C34H34O10	59484555	1937045	0	0	4139901	50246559	2508607	69260852	2.82E+08	0	43993539
8	16	585.2692	8.197227246	15	0	D	C32H40O10	57059457	24883537	0	1295131	23654689	0	0	5398647	0	0	4478854
9	17	541.207	4.297588633	13	0	D	C29H32O10	51765693	0	0	3.93E+08	0	0	0	0	0	0	0
10	21	585.3421	13.57017769	10	0	D	C34H48O8	40379331	0	0	0	0	0	0	0	0	0	0
11	25	637.3368	12.87550885	10	0	D	C37H48O9	37207731	6938174	0	0	11453971	0	0	18376942	0	0	0
12	31	649.2999	10.17540788	16	0	D	C37H44O10	32041569	3050398	0	0	0	0	0	0	0	0	12813287
13	39	591.2585	8.793285861	13	0	D	C34H38O9	24872914	8167630	6593211	0	4638095	2072423	0	17605572	18568763	1593591	5.45E+08
14	40	559.2173	2.5230582	22	0	D	C29H34O11	24089155	0	0	0	0	0	0	0	0	0	0
15	47	487.2325	4.31315775	11	0	D	C27H34O8	19491187	5766567	0	0	0	2126725	0	0	18743957	0	0

Figure 11. Output results of Step 2: Adduct Ion Calculate.

Step 3: Adduct Ion Calculate

Figure 12. Parameters in Step 3: Adduct Ion Calculate.

- (1) Upload CSV Files: Select **xxx-mf.csv** files processed by **Step 2**.
- (2) **Alternatively upload** known compound MS database (**xxx.csv**) in positive or negative ion modes with the following format for MS1 molecular formula matching:

For example, **Thymeleaceae daphnane database.csv**, which included the **molecular formula (M)** and **composition after possible adducts addition (M+Adducts)** of known daphnanes isolated from Thymeleaceae family (until manuscript submitted) was imported.

***Note:** If users do not require comparison with the established compound molecular formula database, uploading database here was unnecessary. As for the positive ion mode, the composition formula after the addition of the adduct should be entered in the form of *ex.* **M+NH₃**, rather than **M+NH₃+H**, due to the charge = +1, and a positive charge would be automatically added during calculation.

	A	B	C	D	E	F
1	Compound Name	Type	M	M+NH ₃	M+CH ₃ NH ₂	M+...
2	resiniferonol	D	C20H28O6	C20H31O6N1	C21H33O6N	...
3	daphneresiniferin A	D	C24H30O8	C24H33O8N1	C25H35O8N	...
4	daphnegen B	D	C24H30O10	C24H33O10N1	C25H35O10N	...
5	daphnediterp A	D	C25H36O9	C25H39O9N1	C26H41O9N	...
6	daphnetoxin (S1)	D	C27H30O8	C27H33O8N1	C28H35O8N	...
7	daphne factor F4	D	C27H32O8	C27H35O8N1	C28H37O8N	...
8	orthobenzoate 2	D	C27H34O8	C27H37O8N1	C28H39O8N	...
9	12-OH-daphnetoxin	D	C27H30O9	C27H33O9N1	C28H35O9N	...
10	daphnegiraldigin	D	C27H32O9	C27H35O9N1	C28H37O9N	...

Figure 13. The .csv format of Thymeleaceae daphnane database.

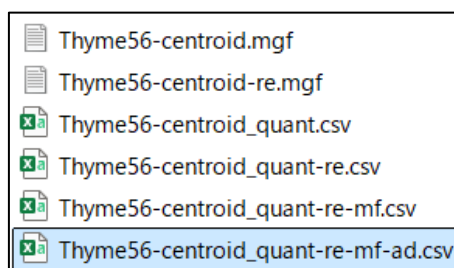
- (2) Define Parameters: Set up charge and mass tolerance.
 - Charge: **+1** (positive ion mode)
 - Mass Tolerance (ppm): **5** (Q-Exactive hybrid quadrupole Orbitrap mass spectrometer)
- (3) Adduct Ion Calculation: Click **"Calculate"** to calculate adduct ion species, and the python

console would be displayed the following information:

```
Run CNPs-MFSA_Web x
Calculated m/z: 615.31637413759, m/z range: (615.3132975557193, 615.3194507194606)
Calculating m/z range for formula: C34H49O10N1, charge: 1, tolerance: 5 ppm
Calculated m/z: 632.3429232386001, m/z range: (632.3397615239838, 632.3460849532163)
Calculating m/z range for formula: C35H50O9, charge: 1, tolerance: 5 ppm
Calculated m/z: 615.35275964631, m/z range: (615.3496828825117, 615.3558364101083)
Calculating m/z range for formula: C35H53O9N1, charge: 1, tolerance: 5 ppm
Calculated m/z: 632.37930874732, m/z range: (632.3761468507763, 632.3824706438637)
Calculating m/z range for formula: C36H40O9, charge: 1, tolerance: 5 ppm
Calculated m/z: 617.27450932561, m/z range: (617.2714229530634, 617.2775956981566)
Calculating m/z range for formula: C36H43O9N1, charge: 1, tolerance: 5 ppm
```

Figure 14. Screenshot of the python console while running Step 3: Adduct ion calculation.

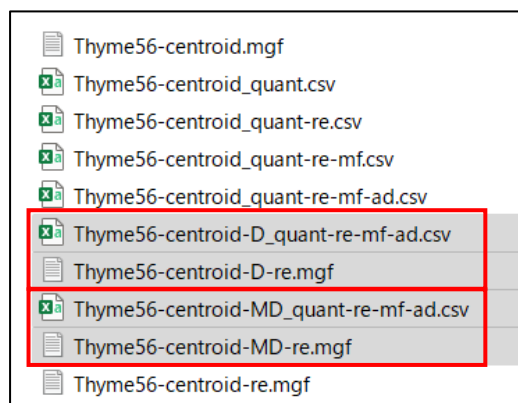
(4) Download Results: download the generated **xxx.zip** file and the results were shown below with new rows, **same peak**, **adduct ion**, **identification**, and **same composition**. The **xxx-ad.csv** was used as the input file and switched to the next tab "Neutral Loss Extract".



	A	B	C	U	V	W	X	Y	Z	AA	AB	AC	AD	AE	AF	AG	AH	BH	BI	BJ	BK												
	row ID	row m/z	row retent	Dod-fr	Pei	Dod-b	Pei	Dod-l	Pea	Dod-r	Pea	Dod-s	Pei	Dje-fr	Pea	Dje-ft	Pea	Dge-fr	Pei	Dps-fr	Pei	Dps-ft	Pei	Dpo-fr	Pea	Dpo-s	Pei	Wdo-w	Pe	D Score	MD Score	type	elemental composition
1	1	655.2525	8.734482	0	0	0	0	0	0	0	0	0	0	8.47E+08	0	0	1.09E+08	0	0	0	0	0	0	0	0	0	15	0	0	0	D	C38H38O10	
2	2	591.2585	8.815949	0	0	0	674910.3	8809730	1687892	0	5.76E+08	0	0	2.44E+09	0	0	0	0	0	0	0	0	0	0	6439611	20	0	0	0	0	D	C34H38O9	
3	3	657.2678	8.894676	0	0	0	0	0	0	0	974000000	0	0	1.18E+08	0	0	0	0	0	0	0	0	0	0	0	0	19	0	0	0	D	C38H40O10	
4	5	649.3003	10.66597	0	806474.1	0	1388519	0	0	0	544000000	7030045	0	3810075	0	0	0	0	0	0	0	0	0	0	0	0	16	0	0	0	D	C37H44O10	
5	6	629.2382	7.71742	0	0	0	4278957	5021816	0	584000000	7110380	0	0	0	0	0	0	0	0	0	0	0	0	0	0	18	0	0	0	0	D	C36H36O10	
6	7	647.2847	10.00882	1664660	0	0	0	0	194000000	0	626000000	0	0	9713001	0	0	0	0	0	0	0	0	0	0	0	0	13	0	0	0	0	D	C37H42O10
7	8	643.2881	10.84486	0	0	0	0	0	0	0	1.13E+08	0	0	1.71E+09	0	0	0	0	0	0	0	0	0	0	0	0	12	0	0	0	0	D	C20H50O22
8	9	603.2223	6.755403	0	0	0	0	0	4851743	0	441000000	2.54E+08	0	0	0	0	0	0	0	0	0	0	0	0	0	0	17	0	0	0	0	D	C34H34O10
9	10	637.3366	12.58025	0	8365642	0	1542323	0	0	0	474000000	0	0	2.65E+08	0	0	0	0	0	0	0	0	0	0	0	0	19	0	0	0	0	D	C37H48O9
10	11	591.2589	7.810607	0	0	0	245751.8	0	0	0	292000000	0	0	1.68E+08	0	0	0	0	0	0	0	0	0	0	0	0	13	0	0	0	0	D	C34H38O9
11	12	781.3587	13.76043	0	0	0	0	0	0	0	212000000	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	11	0	0	0	0	D	C46H52O11

Figure 15. Output results of Step 3: Adduct Ion Calculate.

***Note:** Since the parameters set for type D and type MD in the subsequent workflow were different, the query data need to be split into two data sets (Thyme56-centroid-D_quant-re-mf-ad.csv / Thyme56-centroid-D-re.mgf and Thyme56-centroid-MD_quant-re-mf-ad.csv / Thyme56-centroid-MD-re.mgf) using the **mgf-sep.py** file, which could be found in Github. The above datasets should **run separately** in the subsequent workflow.



Step 4: Neutral Loss Labeling

***Notes:** the "Neutral Loss Extract" tab was designed for the accurate calculation of neutral losses within certain carbon ranges. If users need to calculate normal neutral losses among all product ion pairs or between the precursor ion and product ions, the corresponding .py file [normal neutral losses.py](#) could be found in Github.

Figure 16. Parameters in Step 4: Neutral Loss Labeling.

(1) Upload MGF Files: Select [xxx-re.mgf](#) files processed by **Step 1**.

(2) Upload CSV Files: Select [xxx-ad.csv](#) files processed by **Step 3**.

Upload DB Files: Select [Positive Ion Formula NL.db](#) files generated by [Formula DB Generation.py](#) as **Step 2**.

(3) Define Parameters for type D:

a. Set up charge and mass tolerance.

- Charge: **+1** (positive ion mode)

- Mass Tolerance (ppm): **5** (Q-Exactive hybrid quadrupole Orbitrap mass spectrometer)

b. Set up detailed losses and carbon range:

Target Loss 1: C10 loss: $C_{10}H_{20}O_2$, $C_{10}H_{10}O_4$

(# Corresponding to the substructure of C10 acyl substituents of daphnanes)

Carbon Range 1 From C30 to C20

Carbon Range 2 From C27 to C17

Carbon Range 3 From C37 to C27

Target Loss 2: C11 loss: $C_{11}H_{22}O_2$, $C_{11}H_{10}O_2$

(# Corresponding to the substructure of C11 acyl substituents of daphnanes)

Carbon Range 1 From C31 to C20

Carbon Range 2 From C38 to C27

Carbon Range 3 From C32 to C22

Target Loss 3: C9 loss: $C_9H_8O_2$, $C_9H_8O_3$, $C_9H_{18}O_2$

(# Corresponding to the substructure of C9 acyl substituents of daphnanes)

Carbon Range 1 From C29 to C20

Carbon Range 2 From C36 to C27

Carbon Range 3 From C39 to C30

Target Loss 4: C3 loss: $C_3H_4O_2$

(# Corresponding to the substructure of epoxy moiety of B ring in daphnanes)

Carbon Range 1 From C30 to C27

Carbon Range 2 From C20 to C17

Carbon Range 3 From C27 to C24

Target Loss 5: C2 loss: $C_2H_4O_2$

(# Corresponding to the substructure of acyl substituents of daphnanes)

Carbon Range 1 From C32 to C30

Carbon Range 2 From C22 to C20

Carbon Range 3 From C36 to C34

Target Loss 6: C7 loss: $C_7H_6O_2$

(# Corresponding to the substructure of benzoyl substituents of daphnanes)

Carbon Range 1 From C37 to C30

Carbon Range 2 From C27 to C20

Carbon Range 3 From C29 to C22

Loss Name 1:

Target Loss 1 :

Carbon Range 1 From : **to**

Carbon Range 2 From : **to**

Carbon Range 3 From : **to**

Loss Name 2:

Target Loss 2 :

Carbon Range 1 From : **to**

Carbon Range 2 From : **to**

Carbon Range 3 From : **to**

Loss Name 3:

Target Loss 3 :

Carbon Range 1 From : **to**

Carbon Range 2 From : **to**

Carbon Range 3 From : **to**

Loss Name 4:

Target Loss 4 :

Carbon Range 1 From : **to**

Carbon Range 2 From : **to**

Carbon Range 3 From : **to**

Loss Name 5:

Target Loss 5 :

Carbon Range 1 From : **to**

Carbon Range 2 From : **to**

Carbon Range 3 From : **to**

Loss Name 6:

Target Loss 6 :

Carbon Range 1 From : **to**

Carbon Range 2 From : **to**

Carbon Range 3 From : **to**

Figure 17. Parameters for type D in Step 4: Neutral Loss Labeling.

(4) Define Parameters for type MD:

a. Set up charge and mass tolerance.

- Charge: **+1** (positive ion mode)

- Mass Tolerance (ppm): **5** (Q-Exactive hybrid quadrupole Orbitrap mass spectrometer)

b. Set up detailed losses and carbon range:

Target Loss 1: **C10 loss:** $C_{10}H_{18}O_3$, $C_{10}H_{18}O_2$, $C_{10}H_{16}O_2$, $C_{10}H_{14}O_2$, $C_{10}H_{16}O_3$

(# Corresponding to the substructure of C10 macrocyclic ring of daphnanes)

Carbon Range 1 From C30 to C20

Carbon Range 2 From C27 to C17

Carbon Range 3 From C37 to C27

Target Loss 2: **C14 loss:** $C_{14}H_{24}O_2$, $C_{14}H_{24}O_4$

(# Corresponding to the substructure of C14 macrocyclic ring of daphnanes)

Carbon Range 1 From C34 to C20

Carbon Range 2 From C36 to C22

Carbon Range 3 From C30 to C16

Target Loss 3: **C9 loss:** $C_9H_{16}O_2$

(# Corresponding to the substructure of C9 macrocyclic ring of daphnanes)

Carbon Range 1 From C29 to C20

Carbon Range 2 From C36 to C23

Carbon Range 3 From C31 to C22

Target Loss 4: **C3 loss:** $C_3H_4O_2$

(# Corresponding to the substructure of epoxy moiety of B ring in daphnanes)

Carbon Range 1 From C30 to C27

Carbon Range 2 From C20 to C17

Carbon Range 3 From C34 to C31

Target Loss 5: **C2 loss:** $C_2H_4O_2$

(# Corresponding to the substructure of acyl substituents of daphnanes)

Carbon Range 1 From C32 to C30

Carbon Range 2 From C22 to C20

Carbon Range 3 From C36 to C34

Target Loss 6: C7 loss: $C_7H_6O_2$

(# Corresponding to the substructure of benzoyl substituents of daphnanes)

Carbon Range 1 From C37 to C30

Carbon Range 2 From C29 to C22

Carbon Range 3 From C27 to C20

Loss Name 1:	C10 loss	
Target Loss 1 :	C10H18O3, C10H18O2, C10H16O2, C10H14O2, C10H16O3	
Carbon Range 1 From :	C30	to C20
Carbon Range 2 From :	C27	to C17
Carbon Range 3 From :	C37	to C27 ✖
Loss Name 2:	C14 loss	
Target Loss 2 :	C14H24O2, C14H24O4	
Carbon Range 1 From :	C34	to C20
Carbon Range 2 From :	C36	to C22
Carbon Range 3 From :	C30	to C16 ✖
Loss Name 3:	C9 loss	
Target Loss 3 :	C9H16O2	
Carbon Range 1 From :	C29	to C20
Carbon Range 2 From :	C36	to C23
Carbon Range 3 From :	C31	to C22 ✖
Loss Name 4:	C3 loss	
Target Loss 4 :	C3H4O2	
Carbon Range 1 From :	C30	to C27
Carbon Range 2 From :	C20	to C17
Carbon Range 3 From :	C34	to C31 ✖

Figure 18. Parameters for type MD in Step 4: Neutral Loss Labeling.

Loss Name 5:

Target Loss 5 :

Carbon Range 1 From : to

Carbon Range 2 From : to

Carbon Range 3 From : to

Loss Name 6:

Target Loss 6 :

Carbon Range 1 From : to

Carbon Range 2 From : to

Carbon Range 3 From : to

Loss Name 7:

Target Loss 7 :

Carbon Range 1 From : to

Carbon Range 2 From : to

Carbon Range 3 From : to

Loss Name 8:

Target Loss 8 :

Carbon Range 1 From : to

Carbon Range 2 From : to

Carbon Range 3 From : to

Figure 19. Parameters for type MD in Step 4: Neutral Loss Labeling (continued).

(4) Generate Extraction: Click **"Run Neutral Loss Extraction"** to label target neutral losses within certain carbon range, and the python console would be displayed the following information:

```

Run CNPs-MFSA_Web
Differences: {'C': 10, 'H': 32, 'O': 3}, Target: {'C': 10, 'H': 18, 'O': 2}
Checking target loss - Differences: {'C': 10, 'H': 32, 'O': 3}, Target: {'C': 10, 'H': 16, 'O': 2}
Checking target loss - Differences: {'C': 10, 'H': 32, 'O': 3}, Target: {'C': 10, 'H': 14, 'O': 2}
Checking target loss - Differences: {'C': 10, 'H': 32, 'O': 3}, Target: {'C': 10, 'H': 12, 'O': 2}
Checking target loss - Differences: {'C': 10, 'H': 32, 'O': 3}, Target: {'C': 10, 'H': 10, 'O': 2}
Difference between C27H60O8 and C17H16O0: {'C': 10, 'H': 44, 'O': 8}
Checking target loss - Differences: {'C': 10, 'H': 44, 'O': 8}, Target: {'C': 10, 'H': 18, 'O': 2}
Checking target loss - Differences: {'C': 10, 'H': 44, 'O': 8}, Target: {'C': 10, 'H': 16, 'O': 2}
Checking target loss - Differences: {'C': 10, 'H': 44, 'O': 8}, Target: {'C': 10, 'H': 14, 'O': 2}
Checking target loss - Differences: {'C': 10, 'H': 44, 'O': 8}, Target: {'C': 10, 'H': 12, 'O': 2}

```

Figure 20. Screenshot of the python console while running Step 4: Neutral Loss Labeling.

(5) Export Outputs: download the generated **xxx.zip** file and the results were shown below with new rows, such as **C10 loss**, **C14 loss**, and so on. The **xxx-nl.csv** was used as the input file and switched to the next tab "Feature Ion Extract".

	Thyme56-centroid.mgf
	Thyme56-centroid-D-re.mgf
	Thyme56-centroid-MD-re.mgf
	Thyme56-centroid-re.mgf
	Thyme56-centroid_quant.csv
	Thyme56-centroid_quant-re.csv
	Thyme56-centroid_quant-re-mf.csv
	Thyme56-centroid_quant-re-mf-ad.csv
	Thyme56-centroid-D_quant-re-mf-ad.csv
	Thyme56-centroid-D_quant-re-mf-ad-nl.csv
	Thyme56-centroid-MD_quant-re-mf-ad.csv
	Thyme56-centroid-MD_quant-re-mf-ad-nl.csv

	A	B	C	D	E	F	G	H	I	J	K	BP	BQ	BR	BS	BT	BU	BV
1	row ID	row m/z	row retent	D Score	MD Score	type	same	pea	elemental	adduct	ion	identificat	same	com				
2	1418	680.2911	1.1	8	0	D	none	C33H45O1	[M+NH4]	possibly u	none	none	none	none	C3H4O2	none	none	C7H6O2
3	444	512.2862	1.2	5	0	D	none	C26H41O5	[M+NH4]	possibly k	daphnedi	none	none	none	C3H4O2	none	none	none
4	9757	682.4177	1.2	0	4	MD	none	C36H59O1	[M+NH4]	possibly u	none	none	none	none	none	none	none	none
5	3143	682.4152	1.3	0	5	MD	none	C36H59O1	[M+NH4]	possibly u	none	none	none	none	none	none	none	none
6	10205	469.2223	1.5	6	0	D	10193, 10	C27H32O7	[M+H-H2]	possibly u	none	none	none	none	C3H4O2	none	none	C7H6O2
7	10193	487.2324	1.5	7	0	D	10193, 10	C27H34O8	[M+H]+	possibly k	orthobenz	none	none	none	C3H4O2	none	none	C7H6O2
8	1389	680.2908	1.6	13	0	D	none	C33H45O1	[M+NH4]	possibly u	none	none	none	none	C3H4O2	none	none	none
9	8407	329.1743	1.7	6	0	D	none	C20H24O4	[M+H]+	possibly u	none	none	none	none	none	none	none	none
10	7378	374.22	1.7	5	0	D	none	C19H35O6	[M+NH4]	possibly u	none	none	none	none	C3H4O2	none	none	none
11	10189	451.2111	1.7	6	0	D	none	C27H30O6	[M+H]+	possibly u	none	none	none	none	C3H4O2	none	none	C7H6O2
12	10183	522.2697	1.7	6	0	D	none	C27H39O5	[M+NH4]	possibly k	neogenkw	none	none	none	C3H4O2	none	none	C7H6O2
13	3182	359.1484	1.9	5	0	D	none	C20H22O6	[M+H]+	possibly u	none	none	none	none	none	none	none	none
14	8456	505.2432	2	11	0	D	none	C27H36O5	[M+H]+	possibly k	neogenkw	none	none	none	C3H4O2	none	none	none
15	10405	521.2385	2	7	0	D	none	C27H36O1	[M+H]+	possibly u	none	none	none	none	C3H4O2	none	none	none

Figure 21. Output results of Step 4: Neutral Loss Labeling.

Step 5: Feature Ion Labeling

Figure 22. Parameters in Step 5: Feature Ion Labeling.

- (1) Upload MGF Files: Select **xxx-re.mgf** files processed by **Step 1**.
- (2) Upload CSV Files: Select **xxx-nl.csv** files processed by **Step 4**.
- (3) Define Parameters: Input the diagnostic ion or ion set, and set up detailed **Mass range**, **Charge**, **Mass Tolerance (ppm)**, **Intensity Normalization**, and **Top Number** as following:

***Notes:**

- **Intensity Normalization > 0.1** was used to exclude product ions with relative intensities exceeding 10% within a certain mass range.
- **Top Number N** (n=1-10) represented the top N product ions of extraction intensity from highest to lowest. By modifying **Top Number N**, candidate diagnostic ion can be obtained according to the intensity of the ions, which could use for the generation of candidate structure in the subsequent "annotation" tab.

- Define Parameters for type D:

Ion Name 1: C15 ion

Specific Ion Set 1: C15H18O2

- mass range: 200-260; charge: 1; mass tolerance (ppm): 5; intensity normalization: 0.1; top number: 1.

Ion Name 2: C17 ion

Specific Ion Set 2: C17H24O2, C17H24O3, C17H16O2, C17H18O2, C17H20O2, C17H16O3, C17H18O3, C17H20O3, C17H16O1, C17H18O1, C17H18O4, C17H20O1, C17H22O1, C17H22O2, C17H14O2, C17H14O3, C17H14O1

- mass range: 230-280; charge: 1; mass tolerance (ppm): 5; intensity normalization: 0.05; top

number: 3

Ion Name 3: C18 ion

Specific Ion Set 3: C18H18O2, C18H20O2, C18H22O2, C18H18O3, C18H20O3, C18H22O3, C18H18O1, C18H20O1, C18H22O1, C18H26O1, C18H26O2, C18H24O2, C18H22O1, C18H20O1, C18H201, C18H18O1, C18H18, C18H16O2, C18H16O1, C18H16

- mass range: 200-400; charge: 1; mass tolerance (ppm): 5; intensity normalization: 0.1; top number: 3

Ion Name 4: C19 ion

Specific Ion Set 4: C19H28O3, C19H28O4, C19H22O2, C19H24O2, C19H26O2, C19H22O3, C19H24O3, C19H26O3, C19H22O1, C19H24O1, C19H26O1, C19H20O4, C19H22O4, C19H24O4, C19H20O5, C19H22O5, C19H24O5, C19H20O3, C19H20O2, C19H18O2, C19H18O3, C19H18O1, C19H20O1, C19H20O1, C19H22, C19H24, C19H18O4, C19H16O2, C19H16O3, C19H16O1, C19H16, C19H18, C19H20O1, C19H18O1, C19H16O1

- mass range: 240-340; charge: 1; mass tolerance (ppm): 5; intensity normalization: 0.1; top number: 3

Ion Name 5: C20 ion

Specific Ion Set 5: C20H22O4, C20H22O3, C20H20O3, C20H20O5, C20H20O6, C20H16O2, C20H18O2

- mass range: 200-400; charge: 1; mass tolerance (ppm): 5; intensity normalization: 0.1; top number: 3

Ion Name 6: Acyl A ion

Specific Ion Set 6:

C2H2O, C3H4O, C4H4O, C4H6O, C5H6O, C5H8O, C6H6O, C6H8O, C6H10O, C7H4O, C7H8O, C7H10O, C7H12O, C8H8O, C8H10O, C7H8O2, C8H12O, C8H14O, C9H6O, C9H10O, C8H8O2, C9H12O, C8H10O2, C9H14O, C9H16O, C9H6O2, C10H12O, C9H10O2, C9H10O2, C10H14O, C9H12O2, C10H16O, C8H10O3, C10H18O, C11H8O, C10H10O2, C11H14O, C10H12O2, C11H16O, C10H14O2, C11H18O, C9H12O3, C11H20O, C11H12O2, C12H16O, C11H14O2, C12H18O, C12H20O, C10H14O3, C12H22O, C13H18O, C13H20O, C11H14O3, C13H22O, C11H16O3, C13H24O

- mass range: 50-200; charge: 1; mass tolerance (ppm): 5; intensity normalization: 0.01; top

number: 2

Ion Name 7: Acyl B ion

Specific Ion Set 7: C₁₄H₂₀O, C₁₄H₂₂O, C₁₂H₁₆O₃, C₁₄H₂₄O, C₁₂H₁₈O₃, C₁₄H₂₆O, C₁₅H₂₂O, C₁₅H₂₄O, C₁₅H₂₆O, C₁₅H₂₈O, C₁₆H₂₄O, C₁₆H₂₆O, C₁₆H₂₆O, C₁₆H₂₈O, C₁₆H₃₀O, C₁₇H₂₆O, C₁₇H₂₈O, C₁₇H₂₈O, C₁₇H₃₀O, C₁₇H₃₂O, C₁₈H₂₈O, C₁₈H₃₀O, C₁₈H₃₀O, C₁₈H₃₂O, C₁₈H₃₄O

- mass range: 200-270; charge: 1; mass tolerance (ppm): 5; intensity normalization: 0.01; top number: 2

Ion Name 8: C₅ ion

Specific Ion Set 8: C₅H₄O

- mass range: 50-100; charge: 1; mass tolerance (ppm): 5; intensity normalization: 0.01; top number: 1

Ion Name 9: C₆ ion

Specific Ion Set 8: C₆H₆O

- mass range: 50-100; charge: 1; mass tolerance (ppm): 5; intensity normalization: 0.01; top number: 1

Ion Name 1:	C ₁₅ ion			
Specific Ion Set 1 :	C ₁₅ H ₁₈ O ₂			
Mass Range of Product Ions (m/z):	200	to	260	
Charge:	1	Mass Tolerance (ppm):	5	Intensity Normalization: 0.1 Top Number: 1
Ion Name 2:	C ₁₇ ion			
Specific Ion Set 2 :	C ₁₇ H ₂₄ O ₂ , C ₁₇ H ₂₄ O ₃ , C ₁₇ H ₁₆ O ₂ , C ₁₇ H ₁₈ O ₂ , C ₁₇ H ₂₀ O ₂ , C ₁₇ H ₁₆ O ₃ , C ₁₇ H ₁₈ O ₃ , C ₁₇ H ₂₀ O ₃ , C ₁₇ H ₁₆ O ₁ , C ₁₇ H ₁₈ O ₁ , C ₁₇ H ₁₈ O ₄ , C ₁₇ H ₂₀ O ₁ , C ₁₇ H ₂₂ O ₁ , C ₁₇ H ₂₂ O ₂ , C ₁₇ H ₁₄ O ₂ , C ₁₇ H ₁₄ O ₃ , C ₁₇ H ₁₄ O ₁			
Mass Range of Product Ions (m/z):	230	to	280	
Charge:	1	Mass Tolerance (ppm):	5	Intensity Normalization: 0.05 Top Number: 3
Ion Name 3:	C ₁₈ ion			
Specific Ion Set 3 :	C ₁₈ H ₁₈ O ₂ , C ₁₈ H ₂₀ O ₂ , C ₁₈ H ₂₂ O ₂ , C ₁₈ H ₁₈ O ₃ , C ₁₈ H ₂₀ O ₃ , C ₁₈ H ₂₂ O ₃ , C ₁₈ H ₁₈ O ₁ , C ₁₈ H ₂₀ O ₁ , C ₁₈ H ₂₂ O ₁ , C ₁₈ H ₂₆ O ₁ , C ₁₈ H ₂₆ O ₂ , C ₁₈ H ₂₄ O ₂ , C ₁₈ H ₂₂ O ₁ , C ₁₈ H ₂₀ O ₁ , C ₁₈ H ₂₀ 1, C ₁₈ H ₁₈ , C ₁₈ H ₁₆ O ₂ , C ₁₈ H ₁₆ O ₁ , C ₁₈ H ₁₆			
Mass Range of Product Ions (m/z):	200	to	400	
Charge:	1	Mass Tolerance (ppm):	5	Intensity Normalization: 0.1 Top Number: 3
Ion Name 4:	C ₁₉ ion			
Specific Ion Set 4 :	C ₁₉ H ₂₈ O ₃ , C ₁₉ H ₂₈ O ₄ , C ₁₉ H ₂₂ O ₂ , C ₁₉ H ₂₄ O ₂ , C ₁₉ H ₂₆ O ₂ , C ₁₉ H ₂₂ O ₃ , C ₁₉ H ₂₄ O ₃ , C ₁₉ H ₂₆ O ₃ , C ₁₉ H ₂₂ O ₁ , C ₁₉ H ₂₄ O ₁ , C ₁₉ H ₂₆ O ₁ , C ₁₉ H ₂₀ O ₄ , C ₁₉ H ₂₂ O ₄ , C ₁₉ H ₂₄ O ₄ , C ₁₉ H ₂₀ O ₅ , C ₁₉ H ₂₂ O ₅ , C ₁₉ H ₂₄ O ₅ , C ₁₉ H ₂₀ O ₃ , C ₁₉ H ₂₀ O ₂ , C ₁₉ H ₁₈ O ₂ , C ₁₉ H ₁₈ O ₃ , C ₁₉ H ₁₈ O ₁ , C ₁₉ H ₂₀ O ₁ , C ₁₉ H ₂₂ O ₁ , C ₁₉ H ₂₄ O ₁ , C ₁₉ H ₁₈ O ₄ , C ₁₉ H ₁₆ O ₂ , C ₁₉ H ₁₆ O ₃ , C ₁₉ H ₁₆ O ₁ , C ₁₉ H ₁₆ , C ₁₉ H ₁₈ , C ₁₉ H ₂₀ O ₁ , C ₁₉ H ₁₈ O ₁ , C ₁₉ H ₁₆ O ₁			
Mass Range of Product Ions (m/z):	240	to	340	
Charge:	1	Mass Tolerance (ppm):	5	Intensity Normalization: 0.1 Top Number: 3

Figure 23. Parameters for type D in Step 5: Feature Ion Labeling.

Ion Name 5:

Specific Ion Set 5 :

Mass Range of Product Ions (m/z): to

Charge: Mass Tolerance (ppm): Intensity Normalization: Top Number:

Ion Name 6:

Specific Ion Set 6 :

Mass Range of Product Ions (m/z): to

Charge: Mass Tolerance (ppm): Intensity Normalization: Top Number:

Ion Name 7:

Specific Ion Set 7 :

Mass Range of Product Ions (m/z): to

Charge: Mass Tolerance (ppm): Intensity Normalization: Top Number:

Ion Name 8:

Specific Ion Set 8 :

Mass Range of Product Ions (m/z): to

Charge: Mass Tolerance (ppm): Intensity Normalization: Top Number:

Ion Name 9:

Specific Ion Set 9 :

Mass Range of Product Ions (m/z): to

Charge: Mass Tolerance (ppm): Intensity Normalization: Top Number:

Figure 23 Parameters for type D in Step 5: Feature Ion Labeling (continued).

- Define Parameters for type MD:

Ion Name 1: C10 ion

Specific Ion Set 1: C10H12O3, C10H14O3, C10H16O3

mass range: 110-190; charge: 1; mass tolerance (ppm): 5; intensity normalization: 0.01; top number: 3

Ion Name 2: C14 ion

Specific Ion Set 2: C14H14O3, C14H22O1

mass range: 200-240; charge: 1; mass tolerance (ppm): 5; intensity normalization: 0.1; top number: 2

Ion Name 3: C15 ion

Specific Ion Set 3: C15H22O2, C15H22O3

mass range: 200-260; charge: 1; mass tolerance (ppm): 5; intensity normalization: 0.1; top

number: 2

Ion Name 4: C16 ion

Specific Ion Set 4: C16H16O3

mass range: 230-280; charge: 1; mass tolerance (ppm): 5; intensity normalization: 0.05; top number: 1

Ion Name 5: C17 ion

Specific Ion Set 5: C17H24O2, C17H24O3, C17H16O2, C17H18O2, C17H20O2, C17H16O3, C17H18O3, C17H20O3, C17H16O1, C17H18O1, C17H18O4, C17H20O1, C17H22O1, C17H22O2, C17H14O2, C17H14O3, C17H14O1

mass range: 230-280; charge: 1; mass tolerance (ppm): 5; intensity normalization: 0.05; top number: 3

Ion Name 6: C18 ion

Specific Ion Set 6: C18H18O2, C18H20O2, C18H22O2, C18H18O3, C18H20O3, C18H22O3, C18H18O1, C18H20O1, C18H22O1, C18H26O1, C18H26O2, C18H24O2, C18H22O1, C18H20O1, C18H201, C18H18O1, C18H18, C18H16O2, C18H16O1, C18H16

mass range: 200-400; charge: 1; mass tolerance (ppm): 5; intensity normalization: 0.1; top number: 3

Ion Name 7: C19 ion

Specific Ion Set 7: C19H28O3, C19H28O4, C19H22O2, C19H24O2, C19H26O2, C19H22O3, C19H24O3, C19H26O3, C19H22O1, C19H24O1, C19H26O1, C19H20O4, C19H22O4, C19H24O4, C19H20O5, C19H22O5, C19H24O5, C19H20O3, C19H20O2, C19H18O2, C19H18O3, C19H18O1, C19H20O1, C19H20O1, C19H22, C19H24, C19H18O4, C19H16O2, C19H16O3, C19H16O1, C19H16, C19H18, C19H20O1, C19H18O1, C19H16O1

mass range: 240-340; charge: 1; mass tolerance (ppm): 5; intensity normalization: 0.1; top number: 3

Ion Name 8: C20 ion

Specific Ion Set 8: C20H22O4, C20H22O3, C20H20O3, C20H20O5, C20H20O6, C20H16O2, C20H18O2

mass range: 200-400; charge: 1; mass tolerance (ppm): 5; intensity normalization: 0.05; top number: 3

Ion Name 9: C27 ion

Specific Ion Set 9: C27H32O2, C27H32O3, C27H34O2, C27H34O3, C27H32O4, C27H34O4, C27H32O5, C27H34O5, C27H32O1, C27H34O1, C27H36O3, C27H36O4, C27H36O5, C27H34O6, C27H36O6, C27H36O2, C27H36O1, C27H30O3, C27H30O4, C27H30O5, C27H30O2, C27H32O6

mass range: 350-550; charge: 1; mass tolerance (ppm): 5; intensity normalization: 0.01; top number: 1

Ion Name 10: Acyl A ion

Specific Ion Set 10: C4H6O, C5H8O, C7H4O, C9H6O, C9H6O2

mass range: 50-150; charge: 1; mass tolerance (ppm): 5; intensity normalization: 0.05; top number: 3

Ion Name 1:	C10 ion
Specific Ion Set 1 :	C10H12O3, C10H14O3, C10H16O3
Mass Range of Product Ions (m/z):	110 to 190
Charge:	1
Mass Tolerance (ppm):	5
Intensity Normalization:	0.01
Top Number:	3
Ion Name 2:	C14 ion
Specific Ion Set 2 :	C14H14O3, C14H22O
Mass Range of Product Ions (m/z):	200 to 240
Charge:	1
Mass Tolerance (ppm):	5
Intensity Normalization:	0.1
Top Number:	2
Ion Name 3:	C15 ion
Specific Ion Set 3 :	C15H22O2, C15H22O3
Mass Range of Product Ions (m/z):	200 to 260
Charge:	1
Mass Tolerance (ppm):	5
Intensity Normalization:	0.1
Top Number:	2

Figure 24. Parameters for type MD in Step 5: Feature Ion Labeling.

Ion Name 4: C16 ion

Specific Ion Set 4 :

C16H18O3

Mass Range of Product Ions (m/z): 230 to 280

Charge: 1 Mass Tolerance (ppm): 5 Intensity Normalization: 0.05 Top Number: 1

Ion Name 5: C17 ion

Specific Ion Set 5 :

C17H24O2, C17H24O3, C17H16O2, C17H18O2, C17H20O2, C17H16O3, C17H18O3, C17H20O3, C17H16O1, C17H18O1, C17H18O4, C17H20O1, C17H22O1, C17H22O2, C17H14O2, C17H14O3, C17H14O1

Mass Range of Product Ions (m/z): 230 to 280

Charge: 1 Mass Tolerance (ppm): 5 Intensity Normalization: 0.05 Top Number: 3

Ion Name 6: C18 ion

Specific Ion Set 6 :

C18H18O2, C18H20O2, C18H22O2, C18H18O3, C18H20O3, C18H22O3, C18H18O1, C18H20O1, C18H22O1, C18H26O2, C18H24O2, C18H22O1, C18H20O1, C18H201, C18H18O1, C18H18, C18H16O2, C18H16O1, C18H16

Mass Range of Product Ions (m/z): 200 to 400

Charge: 1 Mass Tolerance (ppm): 5 Intensity Normalization: 0.1 Top Number: 3

Ion Name 7: C19 ion

Specific Ion Set 7 :

C19H28O3, C19H28O4, C19H22O2, C19H24O2, C19H26O2, C19H22O3, C19H24O3, C19H26O3, C19H22O1, C19H24O1, C19H26O1, C19H20O4, C19H22O4, C19H24O4, C19H20O5, C19H22O5, C19H24O5, C19H20O3, C19H22O2, C19H18O2, C19H18O3, C19H18O1, C19H20O1, C19H22O1, C19H22, C19H24, C19H18O4, C19H16O2, C19H16O3, C19H16O1, C19H16, C19H18, C19H20O1, C19H18O1, C19H16O1

Mass Range of Product Ions (m/z): 240 to 340

Charge: 1 Mass Tolerance (ppm): 5 Intensity Normalization: 0.1 Top Number: 3

Ion Name 8: C20 ion

Specific Ion Set 8 :

C20H22O4, C20H22O3, C20H20O3, C20H20O5, C20H20O6, C20H16O2, C20H18O2

Mass Range of Product Ions (m/z): 200 to 400

Charge: 1 Mass Tolerance (ppm): 5 Intensity Normalization: 0.05 Top Number: 3

Ion Name 9: C27 ion

Specific Ion Set 9 :

C27H32O2, C27H32O3, C27H34O2, C27H34O3, C27H32O4, C27H34O4, C27H32O5, C27H34O5, C27H32O1, C27H34O1, C27H36O3, C27H36O4, C27H36O5, C27H34O6, C27H36O6, C27H36O2, C27H36O1, C27H30O3, C27H30O4, C27H30O5, C27H30O2, C27H32O6

Mass Range of Product Ions (m/z): 350 to 550

Charge: 1 Mass Tolerance (ppm): 5 Intensity Normalization: 0.01 Top Number: 1

Ion Name 10: Acyl A ion

Specific Ion Set 10 :

C4H6O, C5H8O, C7H4O, C9H6O, C9H8O2

Mass Range of Product Ions (m/z): 50 to 150

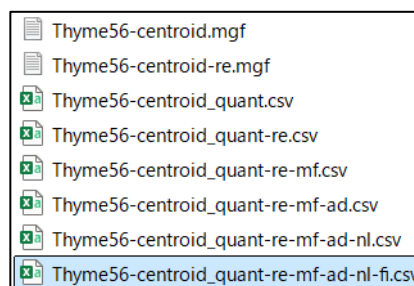
Charge: 1 Mass Tolerance (ppm): 5 Intensity Normalization: 0.05 Top Number: 3

Figure 24. Parameters for type MD in Step 5: Feature Ion Labeling (continued).

***Note:** For detailed ion information, see the [feature ion and neutral loss.txt](#) file, which can be found in GitHub repository [CNP's-MFSA](#).

(4) Generate Extraction: Click "**Run Feature Ion Extraction**" to label target feature ion within certain carbon range.

(5) Export Outputs: download the generated **xxx.zip** file and the results were shown below with new rows, such as **C10 ion**, **C15 ion**, and so on. The **xxx-fi.csv** was used as the input file and switched to the next tab "**Cosine Score Calculation**".



#	A	B	C	D	E	F	G	H	I	J	K	BP	BQ	BR	BS	BT	BU	BV	BW	BX	BY	BZ	CA	CB	CC	CD				
	row ID	row m/z	row retent	D Score	MD Score	type	same	pea	elemental	adduct	ion	identificat	same	com	C10 loss	C14 loss	C9 loss	C3 loss	CO2 loss	C2 loss	C7 loss	C10 ion	C15 ion	C17 ion	C18 ion	C19 ion	C20 ion	C27 ion	Acyl ion	
1	1418	680.2911	1.1	8	0	D	none	C3H4SO	[M+NH4] ⁺	possibly k	daphneditr	none	none	none	none	none	none	none	none	none	none	none	none	none	none	none	none	none	C7H4O1	
2	444	512.2862	1.2	5	0	D	none	C26H41O5	[M+NH4] ⁺	possibly u	none	none	none	none	none	none	none	none	none	none	none	none	none	none	none	none	none	none	none	
3	9757	682.4177	1.2	0	4	MD	none	C36H59O1	[M+NH4] ⁺	possibly u	none	none	none	none	none	none	none	none	none	none	none	none	none	none	none	none	none	none	none	
4	3143	682.4152	1.3	0	5	MD	none	C36H59O1	[M+NH4] ⁺	possibly u	none	none	none	none	none	none	none	none	none	none	none	none	none	none	none	none	none	none	none	
5	10205	469.2223	1.5	6	0	D	10193, 10	C27H32O1	[M+H] ⁺	possibly k	orthobenzi	none	none	none	none	none	none	none	none	none	none	C17H6O2	none	none	none	C19H22O2	C20H22O3	none	C7H4O1	
6	10193	487.2324	1.5	7	0	D	10193, 10	C27H34O1	[M+H] ⁺	possibly u	none	none	none	none	none	none	none	none	none	none	none	C7H6O2	none	none	C17H20O2	none	C19H22O2	C20H22O3	none	C7H4O1
7	1389	680.2908	1.6	13	0	D	none	C33H45O1	[M+NH4] ⁺	possibly u	none	none	none	none	none	none	none	none	none	none	none	none	C17H16O2	C18H20O2	C19H20O3	none	none	none	none	
8	8407	329.1743	1.7	6	0	D	none	C20H24O1	[M+H] ⁺	possibly u	none	none	none	none	none	none	none	none	none	none	none	none	none	C17H16O2	none	C19H22O2	C20H22O3	none	none	none
9	7378	374.22	1.7	5	0	D	none	C19H35O1	[M+NH4] ⁺	possibly u	none	none	none	none	none	none	none	none	none	none	none	C15H22O2	none	C18H22O2	C19H28O4	none	none	C7H8O1, C9H10O2		
10	10189	451.2111	1.7	6	0	D	none	C27H30O1	[M+H] ⁺	possibly u	none	none	none	none	none	none	none	none	none	none	none	C7H6O2	none	none	C18H20O1	C19H22O2	C20H22O3	none	C7H4O1	
11	10183	522.2697	1.7	6	0	D	none	C27H39O5	[M+NH4] ⁺	possibly k	neogekwin	none	none	none	none	none	none	none	none	none	none	C7H6O2	none	none	none	C19H22O2	C20H22O3	none	C7H4O1	
12	3182	359.1484	1.9	5	0	D	none	C20H22O1	[M+H] ⁺	possibly u	none	none	none	none	none	none	none	none	none	none	none	none	none	none	C18H20O2	none	C20H20O5	none	C8H8O2, C10H10O2	
13	8456	505.2432	2	11	0	D	none	C27H36O1	[M+H] ⁺	possibly k	neogekwin	none	none	none	none	none	none	none	none	none	none	none	C17H20O2	C18H22O2	C19H22O3	C20H22O3	none	C7H4O1		
14	10405	521.2385	2	7	0	D	none	C27H36O1	[M+H] ⁺	possibly u	none	none	none	none	none	none	none	none	none	none	none	none	C17H20O3	none	C19H22O3	C20H20O3	none	none	none	
15	10196	531.2224	2.2	8	0	D	none	C28H34O1	[M+H] ⁺	possibly u	none	none	none	none	none	none	none	none	none	none	none	none	none	C18H18	C19H22O3	C20H22O3	none	C7H4O1		
16	484	619.2753	2.2	7	0	D	none	C32H40O1	[M+H] ⁺	possibly u	none	none	none	none	none	none	none	none	none	none	none	C2H4O2	none	none	C17H16O3	C18H18O2	C19H18O3	C20H20O5	none	C5H6O1, C4H6O1
17	6530	477.2119	2.3	7	0	D	6527, 653	C25H32O1	[M+H] ⁺	possibly u	none	none	none	none	none	none	none	none	none	none	none	none	none	C17H16O2	C18H18O2	C19H20O3	none	none	none	none
18	6527	508.2543	2.3	6	0	D	6527, 653	C26H37O1	[M+CH3N]	possibly u	none	none	none	none	none	none	none	none	none	none	none	none	none	C17H16O2	C18H18O2	C19H18O2	none	none	none	none
19	10352	651.3166	2.4	5	0	D	none	C37H46O1	[M+H] ⁺	possibly u	none	none	none	none	none	none	none	none	none	none	none	C7H6O2	none	none	none	none	C19H20O1	C20H22O3	none	C9H6O2, C7H4O1

Figure 25. Output results of Step 5: Feature Ion Labeling.

Step 6: Cosine Score Calculate (Alternative)

1. Target Recognize	2. Composition Calculate	3. Adduct Ion Calculate	4. Neutral Loss Extract	5. Feature Ion Extract	6. Cosine Score Calculate	7. Annotation	8. Visualization
Import Query MGF file (multiple files):		Import Query CSV file (multiple files):		Import Standard MGF file (multiple files):		Import Standard CSV file (multiple files):	
Drop or Select MGF File		Drop or Select CSV File		Drop or Select MGF File		Drop or Select CSV File	
Uploaded MGF files: Thyme56R-re.mgf		Uploaded CSV files: Thyme56R_quant-re-mf-ad-nl-fi.csv		Uploaded MGF files: standard.mgf		Uploaded CSV files: standard_quant.csv	
Mass Range of Product Ions (m/z): 200 to 400							
Intensity Normalization: 0.1 Cosine: 0.7							
Run Cosine Score Calculation Download All Results							
Select file to view results: Select a file							
Please select a file to view results							

- (1) Upload Query MGF Files: Select **xxx-re.mgf** files processed by **Step 1**.
- (2) Upload Query CSV Files: Select **xxx-nl.csv** files processed by **Step 5**.
- (3) Upload Standard MGF Files: Select **standard.mgf** files.
- (4) Upload Standard CSV Files: Select **standard quant.csv** files.

***Notes:** The input standard CSV files should be prepared as the following format with **standards name** and corresponding **substructure type**.

	A	B	C	D	E	F
1	row ID	row m/z	row retention time	peak area	standards name	substructure type
2	1	591.2586	8.3951723	3.95E+09	genkwanine M	6,7-epoxy
3	2	591.2588	8.7453353	2.59E+08	genkwanine N	6,7-epoxy
4	3	533.3109	11.595447	8.44E+08	simplexin	6,7-epoxy
5	4	609.2679	8.3171075	5.39E+09	genkwanine H	6,7-diol
6	5	609.2679	8.0269279	1.30E+07	genkwanine G	6,7-diol
7	6	573.2477	4.1568114	2.67E+08	DPSft27	4,6-epoxy
8	7	649.3006	10.827824	1.59E+09	yuanhuacine	6,7-epoxy
9	8	609.269	3.8598187	7.17E+07	Wlich7 (3.86)	4,7-epoxy
10	9	609.269	6.2090547	8.10E+07	Wlich7 (6.20)	4,7-epoxy

Bring.mgf

ファイル 編集 表示

```

>BEGIN IONS
TITLE=1
PEPMASS=591.2586
SCANS=1
RTINSECONDS=503.71
CHARGE=1+
MSLEVEL=2
52.8127 1.0E5
55.0157 1.0E5
55.2734 1.0E5
62.4463 9.7E4
63.6423 1.1E5

```

- (5) Generate Cosine Score: Click **"Run Cosine Score Calculation"** to calculate.

- (6) Export Outputs: download the generated **xxx.zip** file and the results were shown below with new rows, **cosine score** and **substructure type** (or modified as **B part** for daphnanes). The **xxx-cs.csv** was used as the input file and switched to the next tab "**Annotation**".

- Thyme56R.mgf
- Thyme56R-re.mgf
- Thyme56R_quant.csv
- Thyme56R_quant-re.csv
- Thyme56R_quant-re-mf.csv
- Thyme56R_quant-re-mf-ad.csv
- Thyme56R_quant-re-mf-ad-nl.csv
- Thyme56R_quant-re-mf-ad-nl-fi.csv
- Thyme56R_quant-re-mf-ad-nl-fi-cs.csv

	A	B	C	D	E	F	G	H	I	J	K	BQ	BQ	BQ	BT	BU	BV	BW	BX	CY	BZ	C19	C20	C21	CY	CE	CF									
	row ID	row m/z	row retent.D	Score	MD	Source	type	same	peaks	elemental	adduct	ion	identical	same	CP10	loss	C14	loss	C3	loss	C2	loss	C2	loss	C10	loss	C17	C18	C19	C20	C21	ion	Acyl	ion	cosine score	substructure type
1	1418	680.2911	1.1	8	0	D	none	none	C3H4501[M+NH4] ⁺	possibility u	none	none	none	none	none	none	C14	loss	none	C3H402	none	none	none	C7H602	none	none	C17H1602	C18H1802	C19H2002	none	none	CTH401	none	0.74	6-epoxy	
2	1414	682.2862	1.2	5	0	D	none	none	C2H4141[M+NH4] ⁺	possibility k	daghehnd	none	none	none	none	none	none	none	none	C3H402	none	none	none	none	none	none	none	none	C19H2602	none	none	none	none	value = 0.7	others	
3	9757	682.4177	1.2	0	4	MD	none	none	C3H4591[M+NH4] ⁺	possibility u	none	none	none	none	none	none	none	none	none	none	none	none	none	none	none	none	none	none	none	none	none	none	value = 0.7	others		
4	3143	682.4512	1.3	0	5	MD	none	none	C3H45901[M+NH4] ⁺	possibility u	none	none	none	none	none	none	none	none	none	none	none	none	none	none	none	none	none	none	none	none	none	none	value = 0.7	others		
5	10205	469.2223	1.5	6	0	D	10193, 101	0	C2H7302[M+H+H2] ⁺	possibility u	none	none	none	none	none	none	none	none	none	C3H402	none	none	none	C7H602	none	none	none	none	C19H2202	C20H2202	none	none	CTH401	none	0.83	4-epoxy
6	10193	487.2324	1.5	7	0	D	10193, 101	0	C2H7340[M+H+H2] ⁺	possibility k	trabehn	none	none	none	none	none	none	none	none	C3H402	none	none	none	C7H602	none	none	C17H2002	none	C19H2202	C20H2202	none	none	CTH401	none	0.72	4-epoxy
7	1389	680.2908	1.6	13	0	D	none	none	C3H4501[M+NH4] ⁺	possibility u	none	none	none	none	none	none	none	none	none	C3H402	none	none	none	none	none	none	C17H1502	C18H2002	C19H2402	none	none	none	value = 0.7	others		
8	8407	329.1743	1.7	6	0	D	none	none	C2H0240[M+H+H2] ⁺	possibility u	none	none	none	none	none	none	none	none	none	none	none	none	none	none	none	none	C17H1502	none	C19H2202	C20H2202	none	none	none	value = 0.7	others	
9	10738	374.22	1.7	5	0	D	none	none	C19H3505[M+NH4] ⁺	possibility u	none	none	none	none	none	none	none	none	none	C3H402	none	none	none	none	C15H2202	none	C18H2202	C19H2802	none	none	CTH801	value = 0.7	others			
10	10189	451.2111	1.7	6	0	D	none	none	C2H7300[M+H+H2] ⁺	possibility u	none	none	none	none	none	none	none	none	none	C3H402	none	none	none	C7H602	none	none	C18H2002	C19H2202	C20H2202	none	none	CTH401	value = 0.7	others		
11	10183	522.2697	1.7	6	0	D	none	none	C2H7390[M+H+H2] ⁺	possibility k	neogreen	none	none	none	none	none	none	none	none	C3H402	none	none	none	C7H602	none	none	none	C19H2202	C20H2202	none	none	CTH401	value = 0.7	others		
12	3182	359.1484	1.9	5	0	D	none	none	C2H0220[M+H+H2] ⁺	possibility u	none	none	none	none	none	none	none	none	none	none	none	none	none	none	none	none	C18H2602	none	C20H2202	none	none	CTH802	value = 0.7	others		
13	8456	505.2432	2	11	0	D	none	none	C2H7360[M+H+H2] ⁺	possibility k	neogreen	none	none	none	none	none	none	none	none	C3H402	none	none	none	none	none	none	C17H2002	C18H2202	C19H2202	C20H2202	none	none	CTH401	value = 0.7	others	
14	10405	521.2385	2	7	0	D	none	none	C2H7360[M+H+H2] ⁺	possibility u	none	none	none	none	none	none	none	none	none	C3H402	none	none	none	none	none	none	C17H2002	none	C19H2202	C20H2202	none	none	CTH401	value = 0.7	others	
15	10196	531.2224	2.2	8	0	D	none	none	C2H8401[M+H+H2] ⁺	possibility u	none	none	none	none	none	none	none	none	none	C3H402	none	none	none	C7H602	none	none	none	C18H18	C19H2202	C20H2202	none	none	CTH401	value = 0.7	others	
16	4618	414.2119	2.2	7	0	D	6527, 6533	C2H404	C2H404[M+H+H2] ⁺	possibility u	none	none	none	none	none	none	none	none	none	C3H402	none	none	none	C2H402	none	none	none	none	C18H1802	C19H1802	C20H2002	none	none	CTH501	value = 0.7	6-epoxy
17	6530	477.2119	2.3	6	0	D	6527, 6533	C2H404	C2H404[M+H+H2] ⁺	possibility u	none	none	none	none	none	none	none	none	none	C3H402	none	none	none	none	none	none	none	C17H1502	C18H1502	C19H1502	none	none	CTH501	value = 0.7	6-epoxy	
18	6527	508.2543	2.3	6	0	D	6527, 6533	C2H403	C2H403[M+H+H2] ⁺	possibility u	none	none	none	none	none	none	none	none	none	C3H402	none	none	none	none	none	none	none	C17H1502	C18H1502	C19H1502	none	none	CTH501	value = 0.7	6-epoxy	
19	10352	651.3166	2.4	6	0	D	6527, 6533	C2H401	C2H401[M+H+H2] ⁺	possibility u	none	none	none	none	none	none	none	none	none	C3H402	none	none	none	C2H402	none	none	C17H1502	C18H1502	C19H1502	C20H1502	none	none	CTH402	value = 0.7	4-epoxy	

Step 7: Annotation for Daphnane

***Notes:** The current workflow was based on the following flowcharts. User could modify the source code of "Annotation" tab for module combination of other types of compounds.

(1) Upload CSV Files: Select **xxx-cs.csv** files processed by Step 6.

***Notes:** The column name **substructure type** should be changed to **B part** for daphnane.

(2) Upload DB Files: Select **Skeleton_DB.csv** and **Acyl_DB.csv** files generated by “**pseudo library construction**” files, which could be found in Github. The “pseudo library construction” files contained three parts and can used to generate **Skeleton_DB.csv** and **Acyl_DB.csv**, respectively.

- Acyl Structure Simulation and Feature Prediction
- Skeleton Feature Prediction
- Skeleton Structure Simulation

a). **Skeleton_DB.csv**

	A	B	C	D	E	F
1	Number	A parts	B parts	C parts	M parts	New SMILES
2	1	A1	B1	C1		*OCC120C1C1C30C4(*)OC3(C(=C)C)CC(C)C1(O4)C1C=C(C)C(=O)C1(O)C2O
3	2	A2	B1	C1		*OCC120C1C1C30C4(*)OC3(C(=C)C)CC(C)C1(O4)C1CC(C)C(=O)C1(O)C2O
4	3	A3	B1	C1		*OCC120C1C1C30C4(*)OC3(C(=C)C)CC(C)C1(O4)C1C=C(C)C(O*)C1(O)C2O
5	4	A4	B1	C1		*OCC120C1C1C30C4(*)OC3(C(=C)C)CC(C)C1(O4)C1CC(C)C(O*)C1(O)C2O
6	5	A5	B1	C1		*OCC120C1C1C30C4(*)OC3(C(=C)C)CC(C)C1(O4)C1=CC(C)C(=O)C1(O)C2O
7	6	A1	B1	C2		*OCC120C1C1C30C4(*)OC3(C(=C)C)C(O*)C(C)C1(O4)C1C=C(C)C(=O)C1(O)C2O
8	7	A2	B1	C2		*OCC120C1C1C30C4(*)OC3(C(=C)C)C(O*)C(C)C1(O4)C1CC(C)C(=O)C1(O)C2O
9	8	A3	B1	C2		*OCC120C1C1C30C4(*)OC3(C(=C)C)C(O*)C(C)C1(O4)C1C=C(C)C(O*)C1(O)C2O
10	9	A4	B1	C2		*OCC120C1C1C30C4(*)OC3(C(=C)C)C(O*)C(C)C1(O4)C1CC(C)C(O*)C1(O)C2O
11	10	A5	B1	C2		*OCC120C1C1C30C4(*)OC3(C(=C)C)C(O*)C(C)C1(O4)C1=CC(C)C(=O)C1(O)C2O
12	11	A1	B1	C3		*OCC120C1C1C(O*)C(O)(C(=C)C)CC(C)C1(O)C1C=C(C)C(=O)C1(O)C2O
13	12	A2	B1	C3		*OCC120C1C1C(O*)C(O)(C(=C)C)CC(C)C1(O)C1CC(C)C(=O)C1(O)C2O
14	13	A3	B1	C3		*OCC120C1C1C(O*)C(O)(C(=C)C)CC(C)C1(O)C1C=C(C)C(O*)C1(O)C2O
15	14	A4	B1	C3		*OCC120C1C1C(O*)C(O)(C(=C)C)CC(C)C1(O)C1CC(C)C(O*)C1(O)C2O

b). **Acyl_DB.csv**

	A	B	C	D	E
1	no.	R parts	Acyl SMILES	Acyl Formula	Identification
2	1	R1	CC(=O)*	C2H2O1	acyl
3	2	R2	CCC(=O)*	C3H4O1	propionoyl
4	3	R3	CCCC(=O)*	C4H6O1	butanoyl
5	4	R4	CCCCC(=O)*	C5H8O1	pentanoyl
6	5	R5	CCCCC(=O)*	C6H10O1	hexanoyl
7	6	R6	CCCCCCC(=O)*	C7H12O1	heptanoyl
8	7	R7	CCCCCCCC(=O)*	C8H14O1	octanoyl
9	8	R8	CCCCCCCCC(=O)*	C9H16O1	nonanoyl
10	9	R9	CCCCCCCCC(=O)*	C10H18O1	decanoyl

(3) Generate Annotation: Click "**Run Annotation**" to calculate.

(4) Export Outputs: download the generated **xxx.zip** file and the results were shown below with new rows, such as the **substructure information** and **the combination skeleton and substituents information**. The **xxx-an (index).csv** was used as the input file and switched to the next tab "Visualization".

Thyme56-centroid.mgf
Thyme56-centroid-re.mgf
Thyme56-centroid_quant.csv
Thyme56-centroid_quant-re.csv
Thyme56-centroid_quant-re-mf.csv
Thyme56-centroid_quant-re-mf-ad.csv
Thyme56-centroid_quant-re-mf-ad-nl.csv
Thyme56-centroid_quant-re-mf-ad-nl-fi.csv
Thyme56-centroid_quant-re-mf-ad-nl-fi-cs.csv
Thyme56-centroid_quant-re-mf-ad-nl-fi-cs-an (index).csv
Thyme56-centroid_quant-re-mf-ad-nl-fi-cs-an.csv

substructure information

the combination information

A part	B part	C part	C-12	M part	Acyl part	number	B number	C number	CN	M number	Acyl Number	Acyl SMILES	Skeleton SMILES
1,2-dihydro-3-one	others	9,13,14-orthoester	with C-12	others	5-phenyl-2,4-pentad	A1	others	C2	others	R83,R79	R83,R79	c1ccccc1C=CC=CC	others
1,2-dihydro-3-ol	4,7-epoxy	9,13,14-orthoester	no C-12	others	benzoyl	A4	B7	C1	others	R79	R79	c1ccccc1C(=O)*	others
1,2-dihydro-3-one	others	9,13,14-orthoester	with C-12	others	5-phenyl-2,4-pentad	A2	others	C2	others	R83,R79	R83,R79	c1ccccc1C=CC=CC	others
1,2-en-3-one	6,7-epoxy	9,13,14-orthoester	with C-12	others	decatrienoyl,benzoyl	A1	B1	C2	others	R37,R79	R37,R79	CCCCC=CC=CC(=O)*	*OCC12OC1C1C3O
1,2-en-3-one	others	9,13,14-orthoester	with C-12	others	cinnamoyl,benzoyl	A1	others	C2	others	R80,R79	R80,R79	c1ccccc1C=CC(=O) others	
1,2-en-3-one	others	9,13,14-orthoester	with C-12	others	decatrienoyl,benzoyl	A1	others	C2	others	R48,R79	R48,R79	CCCC=CC=CC=CC(=O) others	
1,2-dihydro-3-ol	others	9,13,14-orthoester	no C-12	others	5-phenyl-2,4-pentad	A4	others	C1	others	R83	R83	c1ccccc1C=CC=CC	others
1,2-en-3-one	others	9,13,14-orthoester	with C-12	others	benzoyl	A1	others	C2	others	R79	R79	c1ccccc1C(=O)*	others
1,2-dihydro-3-ol	6,7-epoxy	9,13,14-orthoester	no C-12	others	decadienoyl	A4	B1	C1	others	R37	R37	CCCCC=CC=CC(=O)*	*OCC12OC1C1C3O
1,2-dihydro-3-ol	6,7-diol	9,13,14-orthoester	no C-12	others	benzoyl	A4	B2	C1	others	R79	R79	c1ccccc1C(=O)*	*OCC1(O)C(O)C2C
1,2-en-3-one	others	9,13,14-orthoester	with C-12	others	5-phenyl-2,4-pentad	A1	others	C2	others	R83,R79	R83,R79	c1ccccc1C=CC=CC	others
1,2-en-3-one	others	9,13,14-orthoester	others	others	benzoyl	others	others	others	others	R79	R79	c1ccccc1C(=O)*	others
1,2-en-3-one	others	9,13,14-orthoester	with C-12	others	5-phenyl-2,4-pentad	A1	others	C2	others	R83,R79	R83,R79	c1ccccc1C=CC=CC	others
1,2-en-3-one	others	9,13,14-orthoester	others	others	acyl,benzoyl	others	others	others	others	R1,R79	R1,R79	CC(=O)*	c1ccccc1C others
1,2-en-3-one	others	9,13,14-orthoester	others	others	5-phenyl-2,4-pentad	others	others	others	others	R83,R79	R83,R79	c1ccccc1C=CC=CC	others
1,2-dihydro-3-one	others	9,13,14-orthoester	with C-12	others	benzoyl	A2	others	C2	others	R79	R79	c1ccccc1C(=O)*	others
1-alkyl-3-ol	6,7-epoxy	9,13,14-orthoester	others	C10 ring with four s	benzoyl	A8	B1	C1	M6	R79	R79	c1ccccc1C(=O)*	*OCC12OC1C1C3O
1-alkyl-3-ol	6,7-epoxy	9,13,14-orthoester	others	C10 ring with one s	benzoyl	A8	B1	C1	M3	R79	R79	c1ccccc1C(=O)*	*OCC12OC1C1C3O
1,2-en-3-one	others	9,13,14-orthoester	with C-12	others	5-phenyl-2,4-pentad	A1	others	C2	others	R83,R79	R83,R79	c1ccccc1C=CC=CC	others
1,2-en-3-one	others	9,13,14-orthoester	with C-12	others	benzoyl	A1	others	C2	others	R79	R79	c1ccccc1C(=O)*	others
1,2-en-3-one	others	9,13,14-orthoester	with C-12	others	acyl,5-phenyl-2,4-pi	A1	others	C2	others	R1,R83	R1,R83	CC(=O)*	c1ccccc1C others
1,2-dihydro-3-one	others	9,13,14-orthoester	with C-12	others	decatrienoyl,benzoyl	A2	others	C2	others	R48,R79	R48,R79	CCCC=CC=CC=CC(=O) others	
1,2-dihydro-3-ol	4,6-epoxy	9,13,14-orthoester	no C-12	others	benzoyl	A4	B8	C1	others	R79	R79	c1ccccc1C(=O)*	others
1-alkyl-3-ol	6,7-epoxy	9,13,14-orthoester	others	C10 ring with three	acyl,benzoyl,nonenc	A8	B1	C1	M5	R1,R79,R23	R1,R79,R23	CC(=O)*	c1ccccc1C *OCC12OC1C1C3O
1,2-en-3-one	others	9,13,14-orthoester	others	others	benzoyl	others	others	others	others	R79	R79	c1ccccc1C(=O)*	others
1,2-en-3-one	others	9,13,14-orthoester	no C-12	others	decadienoyl,hexadi	A1	others	C1	others	R37,R33	R37,R33	CCCCC=CC=CC(=O) others	
1,2-en-3-one	others	9,13,14-orthoester	with C-12	others	cinnamoyl,benzoyl	A1	others	C2	others	R80,R79	R80,R79	c1ccccc1C=CC(=O) others	
1,2-en-3-one	others	9,13,14-orthoester	with C-12	others	benzoyl	A1	others	C2	others	R79	R79	c1ccccc1C(=O)*	others
1,2-en-3-one	others	9,13,14-orthoester	others	others	acyl,benzoyl,decadi	others	others	others	others	R1,R79,R37	R1,R79,R37	CC(=O)*	c1ccccc1C others

(1) Upload CSV Files: Select **xxx-an (index).csv** files processed by **Step 7**.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
1	index	row ID	row m/z	row retent	20221105	20221105	20221207	20221105	20221207	20221105	20221105	20221207	20221105	20221207	20221207	20230217
2	1	1	655.2525	8.7	8.47E+08	1.09E+08	0	0	0	0	0	0	0	0	0	0
3	2	2	591.2585	8.8	5.76E+08	2.44E+09	25100000	68200000	17700000	254145.6	4709378	14700000	2251164	5619186	6663405	26000000
4	3	3	657.2678	8.9	97400000	1.18E+08	0	0	0	0	0	0	0	0	0	0
5	4	5	649.3003	10.7	54400000	3810075	1.04E+08	15400000	12200000	555498.9	16000000	1537602	794179.9	2715672	0	0
6	5	6	629.2382	7.7	58400000	0	0	1.19E+08	70300000	6504685	4538744	4795256	0	0	0	0
7	6	7	647.2847	10	62600000	9713001	18400000	1.31E+09	6.43E+08	1.68E+08	9.33E+08	89600000	0	4.45E+08	0	0
8	7	8	643.2881	10.8	1.13E+08	1.71E+09	676779	396669.9	0	0	488109.4	188933.6	0	0	0	0
9	8	9	603.2223	6.8	44100000	0	60500000	2380547	70200000	0	4175097	2622250	0	0	0	0
10	9	10	637.3366	12.6	47400000	2.65E+08	5415838	0	7999909	0	2830324	1902679	0	2862820	0	7213202

1. Target Recognize

2. Composition Calculate

3. Adduct Ion Calculate

4. Neutral Loss Extract

5. Feature Ion Extract

6. Cosine Score Calculate

7. Annotation

8. Visualization

Import Query CSV file (multiple files):

Drop or Select CSV File

Uploaded CSV files: [Thyme56-centroid_quant-re-mf-ad-nl-fi-cs-an \(index\).csv](#)

Run Visualization

(2) Generate Visualization: Click **"Run Visualization"** to calculate.

- **3D interactive scatter plot** for viewing the overall picture of the plant sample dataset



